

CSE6324 -- Software Verification & Validation

Tutorial & Lab 8

1. What are the benefits of using software test tools and automation?
Speed
 - Efficiency
 - Accuracy and precision
 - Resource reduction
 - Simulation and emulation
 - Relentlessness
2. What are the drawbacks or cautions to consider when deciding to use software test tools and automation?
 - There's no substitute for human eye and intuition
 - Verification is hard to do
 - It's easy to rely on automation too much
 - Some tools are invasive and can cause the software being tested to improperly fail
3. What is one of the simplest, but effective, types of test automation? What features could be added to this test automation to make it even more effective.
One of the simplest yet most effective types of test automation is API smoke testing, which involves sending basic HTTP requests to core endpoints and asserting a successful 200 OK response status. Because it bypasses the complex and often flaky user interface (UI), it executes incredibly fast and provides immediate, reliable feedback on whether the application's backbone is functioning. To make this automation even more effective, we can introduce data-driven testing by linking the scripts to external data sheets, allowing a single test to automatically validate dozens of different input combinations and edge cases. Additionally, integrating these tests into a CI/CD pipeline ensures they trigger automatically with every code commit, transforming a simple uptime check into a powerful, continuous defense against integration bugs.
4. Differentiate between a tool and automation.
A tool is a static instrument or software application used to perform a specific task or facilitate a manual process, whereas automation is the broader system or process of using those tools to execute a sequence of tasks automatically without human intervention. Think of a tool as the *means* and automation as the *execution*; for example, Selenium is a testing tool, but writing a script that uses Selenium to log in, navigate a site, and verify a purchase without a human clicking any buttons is automation. Ultimately, a tool requires continuous human input to achieve an outcome, while automation shifts the labor from the human to the machine by executing predetermined workflows independently.

5. What advantages do smart monkeys have over macros and dumb monkeys?
Smart monkeys knows
 - Where they are
 - What they can do there
 - Where they can go
 - Where they have been
 - If what they are seeing is correct
6. What are the features you can add to a dumb monkey to make it a semi-smart monkey?
 - Add logging to the monkey so that everything it does is recorded to a file
 - Set up a video camera to record what happens to a screen
 - Program the monkey to operate only on the software you are testing
 - Program the monkey to recognize that a crash has occurred, restart the computer, and start running again
7. Under what situation(s) will an interference injector be used during testing?

An interference injector is used during testing to evaluate a system's resiliency, fault tolerance, and recovery mechanisms by deliberately introducing anomalies, delays, or errors into the environment. It is typically deployed in situations where teams need to validate how an application handles real-world chaos, such as severe network degradation (high latency or packet loss), hardware strain (CPU spikes and memory leaks), or corrupted data payloads from failing third-party dependencies. By simulating these hostile conditions, testers can ensure that the system fails gracefully—perhaps by displaying a user-friendly error message or triggering a backup server—rather than suffering a catastrophic, silent crash.